

Audience-based Education: Rethinking the logic of Adaptive Learning Systems Enhanced by Large Language Models

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Abstract—Adaptive Learning Systems (ALSs) have been widely applied to support large-scale personalized learning, yet they continue to face limitations in addressing learners' cognitive, affective, and social learning needs in complex educational contexts. Recent advances in Large Language Models (LLMs) provide new opportunities to rethink the design logic of ALSs through enhanced semantic understanding, generative reasoning, and natural language interaction. Grounded in the Audience-based Education philosophy, this study proposes an LLMs-enhanced adaptive learning system framework that embeds LLMs as an enabling layer within the classic Learner - Domain - Pedagogical architecture. Through coordinated enhancement of learner representation, knowledge organization, and pedagogical decision-making, the proposed framework aims to enrich learners' cognitive, affective, and social learning experiences, thereby improving overall learning effectiveness. By positioning LLMs as an engine rather than a replacement for existing components, this study provides a conceptual basis for understanding how language models can reshape the logic of adaptive learning systems and offers theoretical insights for future research on LLMs-enhanced learning environments.

Keywords—Adaptive Learning Systems, Large Language Models, Personalized Learning

I. INTRODUCTION

Adaptive Learning Systems (ALSs) have been widely developed to support personalized instruction by adjusting learning content, pace, and feedback according to learners' characteristics and learning progress[1]. With advances in artificial intelligence and learning analytics, many existing systems increasingly rely on data-driven methods to infer learners' knowledge states and inform adaptive instructional

decisions[2]. However, the effectiveness of such systems remains constrained by their limited ability to account for learner diversity and the dynamic complexity of real-world educational contexts[3]. In particular, adaptive behaviors in many ALSs are still primarily driven by structured data and predefined representations, which restrict their responsiveness to complex cognitive, affective, and social learning processes.

Recent developments in Large Language Models (LLMs) offer new opportunities to address these limitations. Owing to their strong capabilities in semantic understanding, generative reasoning, and natural language interaction, LLMs enable more flexible interpretation of unstructured learner data and support more context-sensitive instructional responses[4]. These capabilities suggest that LLMs have the potential to extend adaptive learning beyond the constraints of traditional rule-based or task-specific data-driven approaches.

Although growing attention has been paid to integrating LLMs into educational technologies, existing studies largely focus on isolated applications, such as automated feedback generation or intelligent tutoring, rather than rethinking the core architecture of adaptive learning systems[5]. In particular, how LLMs can be systematically embedded into the foundational structure of ALSs to support Audience-based Education goals remains underexplored.

To address this limitation, this study proposes an LLMs-enhanced adaptive learning system framework grounded in a learner-centered educational philosophy. By conceptualizing LLMs as an enabling layer embedded within the classic Learner - Domain - Pedagogical Model[6], the proposed framework aims to enhance learners' cognitive, affective, and social learning experiences within adaptive learning systems, thereby improving overall learning effectiveness. This study provides a conceptual basis for understanding how language

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models can reshape the logic of ALSs and offers theoretical insights for future research on Audience-based Education.

II. RELATED RESEARCH

A. Audience-based Education

The concept of Audience-based Education, which emphasizes adapting teaching to individual needs, was proposed by Confucius based on Confucianism. On the other hand, the Audience-based Education, which places the learner at the heart of the educational process, was advocated by Dewey, who grounded his philosophy in pragmatism and emphasized experience, action, and results. The traditional teaching methods rely on the experience of teachers, which have limitations in terms of the accuracy of students' judgment, the timeliness of academic quality evaluation, and the expansiveness of knowledge. However, technologies like language models not only overcome the aforementioned limitations but also reconstruct the "evaluation-teaching-evaluation" loop, transforming into a more systematic teaching method. ALSs serve as the foundational platform for the convergence of technology and education. Through technology and algorithms, they plan the optimal learning path, embodying the Audience-based Education at their core.

B. Adaptive Learning Systems

ALSs originated from Brusilovsky's adaptive hypermedia framework proposed in 1996 [7]. ALS employ artificial intelligence, data-driven techniques, and machine learning to continuously collect learning evidence, update learner profiles, and adjust instructional strategies, aiming to replace the traditional one-size-fits-all model with personalized learning experiences.

Early ALS were predominantly rule-based, relying on predefined instructional strategies, expert-designed Learner Model, and structured domain representations. Although such systems provided transparent and controllable adaptation, their flexibility and scalability were limited. To realize adaptivity, most ALS adopt a three-model architecture consisting of a Learner Model, a Domain Model, and a Pedagogical Model. The Learner Model is typically implemented using rule-based representations, Bayesian networks, or performance-based statistical models to estimate learners' knowledge states[8]. The Domain Model organizes instructional content through structured knowledge representations such as concept maps or prerequisite graphs. The Pedagogical Model governs instructional decisions using predefined rules, strategy repositories, or optimization mechanisms to select learning activities, feedback, and sequencing strategies[9].

With the increasing availability of learning data, data-driven and machine learning approaches have been introduced to enhance these core models. Statistical modeling and learning analytics enable more fine-grained inference of learner states and adaptive decision-making based on interaction and performance data[10]. However, adaptive behaviors often remain constrained by predefined features, task-specific assumptions, and limited semantic understanding[11][12].

Despite these advances, existing ALSs still face challenges in supporting complex, Audience-based Education goals. In

particular, limited capabilities in interpreting unstructured learning data, modeling affective and social dimensions, and generating context-sensitive pedagogical responses restrict their ability to support Audience-based Education. These challenges motivate the exploration of new adaptive paradigms that integrate semantic understanding, flexible knowledge representation, and interactive pedagogical reasoning.

C. LLMs in Adaptive Learning Systems

Recent advances in LLMs, a large-scale neural models trained on extensive textual data, have significantly expanded the design space of ALSs. Compared with traditional AI techniques used in ALSs, such as rule-based methods or task-specific models, LLMs are characterized by strong semantic understanding, generative capability, and natural language interaction. These properties enable ALSs to move beyond rigid, predefined representations and to process unstructured learning data, generate adaptive instructional content, and support continuous learner – system interaction.

Technically, LLMs provide several affordances that are particularly relevant to adaptive learning. Their semantic representation capability allows systems to infer learners' cognitive states and misconceptions from textual responses and interaction traces [13], rather than relying solely on observable performance indicators. Their generative nature supports the on-demand creation of explanations, examples, and feedback tailored to individual learners [14], reducing dependence on pre-designed instructional resources. In addition, their conversational interaction capability enables dialog-based adaptation [15], facilitating more timely and learner-responsive instructional support [16].

These affordances help address persistent limitations of traditional ALSs, which typically depend on structured data and predefined knowledge representations and therefore struggle to model affective and social aspects of learning [17]. By analyzing learners' language use, emotional expressions, and collaborative dialogue [18], LLMs provide richer inputs for learner modeling and enable more flexible pedagogical strategies [19]. As a result, LLMs offer a promising pathway to balance scalability and personalization in adaptive learning environments. Table 1 compares regular ALSs with LLMs-enhanced ALSs across five dimensions.

TABLE I. COMPARISON BETWEEN REGULAR ALSS AND LLMs-ENHANCED ALSS

Dimension	regular ALSs	LLMs-Enhanced ALSs
System design paradigm	Rule- or model-driven adaptation based on predefined structures	LLM-enabled paradigm integrating semantic understanding and generative reasoning
Learner modeling logic	Primarily cognitive and performance-oriented	Multi-dimensional modeling of cognitive, affective, and social characteristics
Knowledge and content adaptation	Static or semi-dynamic content selection from predefined repositories	Dynamic knowledge organization and content generation supported by LLMs
Pedagogical strategy generation	Pre-authored rules or limited adaptive policies	On-demand strategy generation through LLMs' generative capability

Dimension	regular ALSs	LLMs-Enhanced ALSs
Personalization and interaction	Limited personalization with form-based or one-way interaction	Scalable personalization through continuous, dialog-based interaction

III. ALSs FRAMEWORK BASED ON THREE CORE PILLARS

Building on prior research on ALSs, which commonly adopt a Learner Model, a Domain Model, and a Pedagogical Model as their core components, this study proposes an LLMs-enhanced adaptive learning system framework. By integrating LLMs as an enabling layer within the classic three-model architecture, this framework aims to enhance the learner's representation, knowledge organization, and teaching decision-making execution[20], while retaining the basic structure of traditional ALSs. In this framework, the functional responsibilities of the three models are as follows: the Learner Model focuses on state inference, the Domain Model on knowledge organization, and the Pedagogical Model on decision-making and intervention execution. Aims to strengthen learner representation, knowledge organization, and pedagogical decision-making, while preserving the foundational structure of traditional ALSs[21]. The following subsections describe how LLMs enhance each core model and jointly support Audience-based Education.

A. Learner Model: The Core of Constructing Multi-Dimensional Dynamic Profiles

From the perspective of Audience-based Education, the primary logical starting point of an adaptive learning system lies in a fine-grained understanding of learners. Accordingly, the Learner Model is the core foundation for a system to implement Audience-based Education. It focuses on dynamically characterizing learners' multi-dimensional features and predicting trends[22]. The goal is to overcome the limitation of traditional models that only focus on knowledge level, achieving comprehensive coverage and dynamic evolution of cognitive, affective, and social features.

In terms of system functions, this model consists of three dimensions: cognitive, affective, and social.

1) The cognitive dimension analyzes learners' thinking expressions to accurately assess their knowledge levels and learning styles, and predict their cognitive development trends, providing a basis for personalized knowledge delivery.

2) The affective dimension mines learners' emotional tendencies and intrinsic motivation from learning interaction texts and behavioral data, identifying key emotional nodes that influence learning outcomes.

3) The social dimension analyzes interaction behaviors and dialogue quality in collaborative learning to characterize learners' collaborative abilities and other traits, facilitating peer matching and collaborative task assignment.

Technically, the Learner Model leverages LLMs as a semantic and generative backbone, combined with federated learning mechanisms to ensure privacy-preserving implementation. It safely combines data from multiple platforms while protecting privacy. It also utilizes the "learner digital twin" idea with LLMs to predict future learning states, enabling targeted interventions. The model collects data from answers, interactions, knowledge progress, and collaboration

records. It employs the semantic understanding and conversational analysis capabilities of LLMs to evaluate learners' thinking patterns, emotional expressions, and communication quality. Deep learning and clustering build a dynamic profile covering cognitive, affective, and social features. To enhance accuracy, the model can add physiological and contextual data to create a four-dimensional collection system. LLMs cross-modal alignment breaks data barriers for better integration. For example, linking anxiety peaks with negative text and answer errors improves affective and cognitive modeling. A data quality system filters low-quality data using anomaly detection and credibility grading. This ensures model accuracy. The resulting profile moves beyond traditional knowledge-focused models to provide dynamic, multi-dimensional learner insights.

B. Domain Model: The Dynamic Organizing Foundation of Multi-modal Knowledge

Domain Model structures and dynamically updates knowledge. It moves beyond simple concept links in traditional systems to build a "knowledge-situation-capability" framework, providing precisely matched knowledge for personalized learning.

Functionally, the model operates through three core modules: structured knowledge construction, multi-modal content association, and dynamic knowledge update.

1) The structured knowledge construction module employs LLMs to extract key points from textbooks, courses, and cases. It builds knowledge graphs with concept links, cognitive levels, and application scenarios. It also links knowledge to cognitive abilities and social needs.

2) The multi-modal content association module combines text, audio, video, and virtual labs. It utilizes LLMs to connect knowledge to resources. It provides different content for different learners, like diagrams for visual learners and audio for auditory learners.

3) The dynamic knowledge update module utilizes data from the Learner Model, such as knowledge mastery, emotions, and collaboration. With LLMs, the system updates knowledge graphs and resources. For example, it creates new examples and exercises for difficult topics.

In technical perspective, Domain Model uses ontology to strengthen semantic links between knowledge. By defining core concepts, hierarchical relationships, multi-dimensional attributes, and semantic rules in a subject area, it builds a structured knowledge framework and deepens cross-knowledge logical connections, overcoming limitations of traditional knowledge graphs. It also links multi-modal learning resources to ontology concepts, facilitating precise knowledge matching and intelligent reasoning. Using natural language processing, the model reads uploaded materials like textbooks and cases to extract new knowledge and updates. Combined with learning effect data from the Learner Model, the system dynamically supplements and optimizes knowledge graphs and resources, adjusts knowledge organization and links in real time, and generates personalized resources like targeted examples and exercises. This process enables rapid iteration of the knowledge system, ensuring its timeliness and applicability,

and ultimately builds and continuously enhances the "knowledge-situation-capability" framework.

C. Pedagogical Model: An Execution Engine for Multi-Dimensional Collaborative Strategies

The Pedagogical Model serves as a coordination and execution layer connecting the Learner Model and the Domain Model, with LLMs enabling adaptive decision-making through generative and conversational interaction. With its core lying in the intelligent matching of learners' multi-dimensional profiles with knowledge resources. Pedagogical Model has four modules: cognitive adaptation, emotional intervention, social collaboration, and three-dimensional coordination.

1) The cognitive adaptation module utilizes the learners' cognitive profile to select matching knowledge and paths from Domain Model. It applies reinforcement learning to adjust content difficulty and pace. It also provides personalized cognitive support for knowledge gaps, like using LLMs to generate Socratic questions that guide thinking instead of giving direct answers.

2) The emotional intervention module creates different incentives and adjustments based on the learners' emotional state. For anxious learners, it sends moderately difficult tasks and encouraging feedback. For uninterested learners, it employs LLMs to create fun learning situations and examples to boost motivation.

3) The social collaboration module applies the learners' social role tags to match peers and assign team roles. It employs LLMs to design collaboration scripts for effective communication and knowledge creation. It also analyzes interaction data to evaluate collaboration and improve strategies.

4) The three-dimensional coordination module combines needs from all three dimensions. It builds a collaborative decision-making system to avoid one-sided strategies. For example, when assigning collaboration tasks, it considers cognitive, emotional, and social factors to ensure suitability and effectiveness.

In terms of technology, Pedagogical Model uses LLMs to create a digital tutor. It responds to learners' cognitive, emotional, and social needs through natural interactions like Socratic questioning, personalized explanations, and contextual guidance. Using LLMs' dynamic content generation, the model creates cognitive support matching learners' levels, motivational content for their emotions, and scripts for collaboration. It also applies ontology to strengthen semantic knowledge links, ensuring teaching strategies match knowledge logic. Through cross-platform APIs, it connects with Learning Management Systems and collaboration tools. This enables efficient resource use, automatic task assignment, and structured collaboration support. Real-time data sharing through these APIs supports dynamic strategy adjustments and closed-loop mechanism of the Learner Model, Domain Model, and Pedagogical Model. This allows efficient execution of the "cognitive adaptation - emotional intervention - social collaboration" teaching strategy.

Fig. 1 presents the overall conceptual framework of an LLMs-enhanced adaptive learning system, in which the Learner Model, Domain Model, and Pedagogical Model are organized into a closed-loop mechanism. Within this framework, LLMs function as a cross-cutting cognitive engine that supports semantic understanding, generative reasoning, and interactive coordination across the three core models.

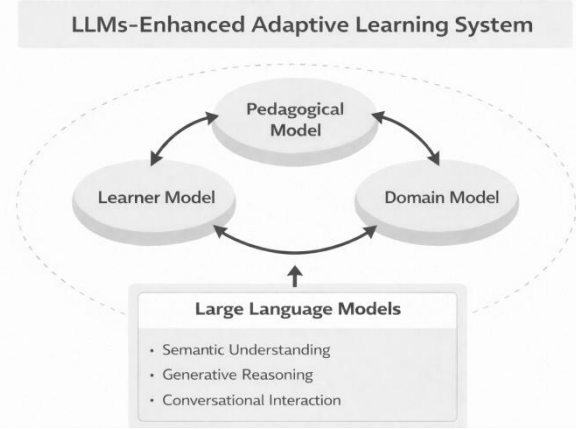


Fig. 1. LLMs-enhanced adaptive learning system framework

D. Four Key Steps to Collaborate Three Models

The core strength of this ALSs lie in the deep collaboration and closed-loop mechanism of its three models, enabled by the semantic understanding, generative capability, and conversational interaction of LLMs. Their interaction follows four key steps: data-driven modeling, precise resource matching, intelligent intervention and feedback optimization.

1) In data-driven modeling, the system captures real-time learner data like answers, interactions, and collaboration records. This data feeds into the Learner Model, which generates a dynamic profile of cognitive level, emotional state, and social traits through multi-dimensional analysis.

2) In precise resource matching, Learner Model shares this profile with the Pedagogical Model. Based on the learners' needs, the Pedagogical Model requests specific knowledge resources from Domain Model, which utilizes its structured knowledge system and multi-modal resources to quickly select matching content, paths, and materials, ensuring a precise link between learner needs and knowledge resources.

3) In intelligent intervention, Pedagogical Model combines selected resources with its integrated teaching strategies, including cognitive adaptation, emotional intervention, social collaboration, to create a personalized learning plan. It delivers this plan to the learners' device and uses a human-like digital tutor for real-time interaction through Socratic questioning and targeted feedback, ensuring an adaptive and effective learning experience.

4) In feedback optimization, learning outcome data flows back to all three models. It updates the Learner Model's profile, improves the Pedagogical Model's strategies, and provides insights for the Domain Model to refine its knowledge graph and resources.

Through a conceptual cycle of data-driven modeling, precise resource matching, intelligent intervention, and feedback optimization, the three models are interrelated and continuously exchange information. These steps illustrate how an LLM-based adaptive learning framework moves beyond basic knowledge adaptation toward supporting learners' cognitive, emotional, and social development, and operationalizes the principles of Audience-based Education by continuously aligning instructional decisions with learners' evolving multi-dimensional needs. Fig. 1 presents the overall structural organization of the proposed framework, while Fig. 2 conceptually depicts this cyclic process from an operational perspective.

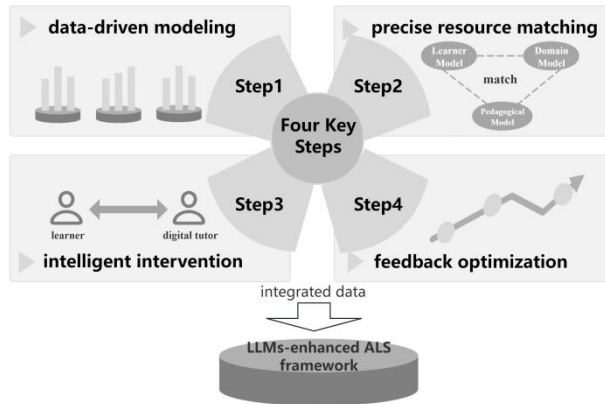


Fig. 2. steps of combining three models

IV. LIMITATIONS AND FUTURE RESEARCH DIRECTIONS

Although this study has proposed a theoretical framework for an adaptive learning system based on LLMs, several limitations remain. In terms of technical implementation, challenges such as stability, controllability, and computational cost persist. Future research should investigate robust strategies, including model calibration, privacy protection mechanisms, and efficiency optimization, to enhance practical feasibility. From a methodological perspective, the framework has not been empirically validated to confirm its effectiveness in supporting personalized learning. Further studies are needed to implement the framework in real educational settings and conduct rigorous evaluations to assess its impact on learners.

V. CONCLUSION

This study reconceptualizes adaptive learning systems from the perspective of Audience-based Education by extending the classic Learner–Domain–Pedagogical framework. It clarifies how the Learner Model, Domain Model, and Pedagogical Model can jointly support individualized learning in ALSs. Within this framework, LLMs are positioned as an enabling layer that enhances existing adaptive mechanisms by strengthening semantic understanding, interactive processes, and pedagogical expressiveness. Rather than replacing established models, LLMs complement and extend their capabilities in supporting diverse and dynamic learning needs. As a conceptual contribution, this study provides a theoretical foundation for understanding how LLMs may reshape the logic of adaptive learning systems and lays the

groundwork for future empirical research on the design, implementation, and evaluation of LLMs-enhanced adaptive learning systems.

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